

Differential Equations as Coalgebras

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We express differential equations as coalgebras, γ_f , and show that their solutions define morphisms to a candidate final coalgebra, ω . For existence and uniqueness of solutions, we need some boundedness assumptions on the functions involved.

Consider a differential equation:

$$\frac{dy}{dt} = f(y),$$

where y is some function of time and we assume that f does not depend on time. Explicitly, $f: \mathbb{R} \rightarrow \mathbb{R}$ and $y: [0, T) \rightarrow \mathbb{R}$, for some $T \in \mathbb{R}$; we denote the composition of these two as $f(y): [0, T) \rightarrow \mathbb{R}$.

Consider the nonstandard extensions $f^*: \mathbb{R}^* \rightarrow \mathbb{R}^*$ and $y^*: [0, T)^* \rightarrow \mathbb{R}^*$, and fix an infinitesimal $dt \in \mathbb{R}^*$. With these, define the coalgebras

$$\begin{aligned} \gamma_f: \mathbb{R}^* &\rightarrow \bar{\mathbb{R}} \times \mathbb{R}^* & \gamma_f(y_0) &:= (y_0^\circ, y_0 + f^*(y_0) \cdot dt) \\ \omega: \mathbb{R}^{*[0, T)^*} &\rightarrow \bar{\mathbb{R}} \times \mathbb{R}^{*[0, T)^*} & \omega(y) &:= (y(0)^\circ, \lambda t. y(t + dt)) \end{aligned}$$

of the functor $F: \mathbf{Set} \rightarrow \mathbf{Set}$ given by $FX := \bar{\mathbb{R}} \times X$.

A solution of the differential equation, parametrised by the initial state, is a function $Y: \mathbb{R} \rightarrow \mathbb{R}^{[0, T)}$ such that, for every initial state $y_0 \in \mathbb{R}$, $Y(y_0)(0) = y_0$ and $Y(y_0)' = Y(y_0) \circ f$.

The rest of this note shows that a solution Y of the differential equation given by f determines a coalgebra morphism $Y^*: \gamma_f \rightarrow \omega$, i.e. $Y^* \circ \omega = \gamma_f \circ (\mathbb{R} \times Y^*)$. We first characterise coalgebra morphisms of type $\gamma_f \rightarrow \omega$.

Lemma 0.1. *A function $s: \mathbb{R}^* \rightarrow \mathbb{R}^{*[0, T)^*}$ is a coalgebra morphism $s: \gamma_f \rightarrow \omega$ if and only if, for all $y_0 \in \mathbb{R}^*$ and all $t \in [0, T)^*$,*

$$s(y_0)(0)^\circ = y_0^\circ \quad \text{and} \quad s(y_0)(t + dt) = s(y_0 + f^*(y_0)dt)(t).$$

Proof. By evaluating pointwise the equation $s \circ \omega = \gamma_f \circ (\mathbb{R} \times s)$. □

Euler approximations provide the unique solutions of differential equations [Kei76, §14A, §14B].

Definition 0.2 (Euler approximations [Kei76, §14A]). Consider an increment $\Delta t \in \mathbb{R}^+$ and the largest natural number $n \in \mathbb{N}$ such that $n \cdot \Delta t < T$. For each $y_0 \in \mathbb{R}$, define the sequence $y_{i+1} := y_i + f(y_i) \cdot \Delta t$, for $i = 1, \dots, n$. The finite Euler approximation of the solution of the differential equation given by f , starting at y_0 and with increment Δt , is the polygonal function that interpolates $y_0, \dots, y_n$.

This procedure defines a function $a_f: \mathbb{R} \times \mathbb{R}^+ \rightarrow \mathbb{R}^{[0, T]}$ that maps an initial state, $y_0 \in \mathbb{R}$, and an increment, $\Delta t \in \mathbb{R}^+$, to the associated finite Euler approximation $a_f(y_0, \Delta t) \in \mathbb{R}^{[0, T]}$ of the solution of the differential equation given by f .

The hyperfinite Euler approximation of the solution of the differential equation given by f is the extension $a_f^*: \mathbb{R}^* \times \mathbb{R}^{+*} \rightarrow \mathbb{R}^{*[0, T]^*}$.

Remark 0.3. By induction one shows that finite Euler approximations satisfy

$$a_f(y_0, \Delta t)(i \cdot \Delta t + \Delta t) = a_f(y_0 + f(y_0) \cdot \Delta t, \Delta t)(i \cdot \Delta t),$$

for all $y_0 \in \mathbb{R}$, all $\Delta t \in \mathbb{R}^+$ and all $i = 0, \dots, n$. Then, the transfer principle gives us that hyperfinite Euler approximations satisfy

$$a_f^*(y_0, dt)(h \cdot dt + dt) = a_f^*(y_0 + f^*(y_0) \cdot dt, dt)(h \cdot dt),$$

for all $y_0 \in \mathbb{R}^*$, all $dt \in \mathbb{R}^{+*}$ and all $h \in \mathbb{N}^*$ such that $h \cdot dt < T$. Then, we obtain that

$$a_f^*(y_0, dt)(t + dt) = a_f^*(y_0 + f^*(y_0) \cdot dt, dt)(t),$$

for all $y_0 \in \mathbb{R}^*$, all $dt \in \mathbb{R}^{+*}$ and all $t \in [0, T]^*$.

We also have that $a_f^*(y_0, dt)(0) = y_0$, for all $y_0 \in \mathbb{R}^*$ and all $dt \in \mathbb{R}^{+*}$.

Theorem 0.4 ([Kei76, Theorems 14.2 and 14.9]). *Fix two positive reals, M and ϵ ; suppose that $|f| \leq M$ and $|y - y_0| \leq M \cdot t + \epsilon$ in the interval $[0, T]$; suppose that f is locally Lipschitz on $\mathbb{R}$. Then, every hyperfinite Euler approximation $a_f^*(y_0, dt)$ has the same standard part that is the unique solution of the initial value problem*

$$\frac{dy}{dt} = f(y) \quad y(0) = y_0$$

in the interval $[0, T]$.

Finally, we show that hyperfinite Euler approximations determine coalgebra morphisms.

Proposition 0.5. *Consider the hyperfinite Euler approximation $a_f^*: \mathbb{R}^* \times \mathbb{R}^{+*} \rightarrow \mathbb{R}^{*[0,T]^*}$ of the solution of the differential equation given by f . Then, the function $Y^* := a_f^*(-, dt)$ is a coalgebra morphism $\gamma_f \rightarrow \omega$.*

Proof. By Remark 0.3, we have that

$$Y^*(y_0)(0) := a_f^*(y_0, dt)(0) = y_0$$

and that

$$\begin{aligned} & Y^*(y_0)(t + dt) \\ &:= a_f^*(y_0, dt)(t + dt) \\ &= a_f^*(y_0 + f^*(y_0) \cdot dt, dt)(t) \\ &=: Y^*(y_0 + f^*(y_0) \cdot dt)(t) \end{aligned}$$

By Lemma 0.1, Y^* is a coalgebra morphism of type $\gamma_f \rightarrow \omega$. □

Future work. We would like the converse to be also true: every coalgebra morphism of type $\gamma_f \rightarrow \omega$ should define a solution of the differential equation associated to f . This would prove that the coalgebra ω is final.

References

- [Kei76] H Jerome Keisler. *Foundations of infinitesimal calculus*, volume 20. Prindle, Weber & Schmidt Boston, 1976.